

Conditional EEG Data Augmentation via Dynamic Topology-aware and Task-synergistic Diffusion Model

Chuncheng Liao^a, Xiangcun Wang^a, Jiakai Zhang^{a,*}

^a*School of Artificial Intelligence, Beijing Normal University, Beijing 100875, China*

Abstract

Motor imagery brain-computer interfaces (MI-BCIs) suffer from limited training data and cross-session signal non-stationarity. This study proposes DTTD, a conditional diffusion framework for EEG data augmentation that generates high-density channel data from sparse-channel inputs guided by task labels. DTTD integrates four modules: a task-conditioned dynamic topology module that adaptively models electrode spatial dependencies, a multi-scale frequency decoupling module with task-adaptive band weights, a gated feature fusion mechanism, and a light-noise training strategy with direct X_0 prediction for efficient single-step inference. Task conditioning enables DTTD to generate class-discriminative samples with distinct ERD patterns, yielding significantly higher inter-class variation than unconditional methods. On BCI Competition IV 2a, DTTD augmentation improves within-subject accuracy to 78.38% (+3.90 pp over the 22-channel reference, $p < 0.01$) and cross-session accuracy to 65.70% (+2.62 pp, $p < 0.01$). On HGD, DTTD exceeds the 128-channel reference by 6.08 pp in cross-subject and 3.53 pp in cross-session scenarios. These results demonstrate that conditional generation producing task-discriminative samples is more effective for classification augmentation than faithful signal reconstruction.

Keywords: EEG data augmentation, diffusion model, topology awareness, task synergy, motor imagery

1. Introduction

Electroencephalogram (EEG), as a direct representation of brain neural activity, holds irreplaceable value in brain-computer interfaces (BCIs), clinical diagnosis, and cognitive neuroscience research [1, 2, 3]. With millisecond-level temporal resolution and non-invasive acquisition characteristics, EEG has become an important modality for analyzing neural mechanisms and constructing human-machine interaction systems. However, motor imagery brain-computer interfaces (MI-BCIs) face two challenges in practical applications. On one hand, the number of EEG training samples per subject is typically limited (tens to hundreds of trials per class), while deep learning classifiers require large amounts of data to adequately learn task-relevant discriminative features, leading to model overfitting and limited generalization capability. On the other hand, EEG signals exhibit significant variability across different recording sessions, influenced by factors such as electrode impedance drift, changes in subject state, and environmental noise [4, 5]. Classifiers trained on one session often experience 5–15% performance degradation on new sessions, severely limiting the long-term usability of BCI systems [6, 7]. Recent works leveraging contrastive networks, spiking neural networks, cross-brain interaction modeling, and large language models have further improved AI applications in medicine and EEG decoding for motor imagery, emotion recognition, and intra-cortical signals [8, 9, 10, 11], yet the fundamental challenges of data scarcity and cross-session non-stationarity remain unresolved.

Data augmentation offers a promising avenue for addressing these challenges. Traditional methods such as spherical spline interpolation [12], Laplacian estimation [13], and kriging [14] assume smooth spatial potential distributions

*Corresponding author.

Email addresses: liaocc3450@bnu.edu.cn (Chuncheng Liao), wangxiangcun@mail.bnu.edu.cn (Xiangcun Wang), jiakai.zhang@bnu.edu.cn (Jiakai Zhang)

but cannot model complex, task-specific neural patterns. While deep learning models like CNNs [15] and autoencoders [16] offer improvements, they still lack adaptability to multi-scale spatiotemporal features, overlook topological relationships between electrodes, and do not integrate task-relevant physiological constraints.

With the development of deep generative models, new pathways have emerged for EEG data augmentation [17, 18]. Variational Autoencoders (VAEs) [19, 20] and Generative Adversarial Networks (GANs) [21, 22] have been applied to EEG generation tasks. However, both dominant generative models have inherent limitations: GANs suffer from training instability and mode collapse, resulting in generated samples lacking diversity; VAEs, due to their Gaussian prior assumption, typically produce over-smoothed samples with limited diversity and lack class discriminability.

Recently, Denoising Diffusion Probabilistic Models (DDPMs) [23] have achieved remarkable success in image and audio generation by learning gradual denoising processes, and have been adapted for EEG synthesis and reconstruction [24, 25, 26, 27]. Score-based models [28] unified diffusion from a continuous-time perspective, and denoising score matching [29] provided efficient training objectives. Conditional diffusion models [30] introduced class labels or text for guided generation, and classifier-free guidance [31] further improved conditional generation quality. These advances make diffusion models a compelling framework for EEG data augmentation: by generating high-density EEG samples from sparse-channel inputs guided by task labels, both the diversity and scale of training data can be expanded while preserving class-discriminative information. Despite these advances, existing EEG reconstruction diffusion methods still exhibit three critical shortcomings. First, they lack dynamic topology modeling capability. Methods such as EEGDiff and BrainDiff rely on static electrode adjacency matrices based on physical distances or predefined functional connectivity templates, which cannot adapt to task-dependent neural dynamics. During motor imagery, the contralateral motor cortex activates while sensorimotor connectivity dynamically reconfigures [32]. Static adjacency matrices cannot capture such task-specific changes in spatial dependencies, resulting in compromised spatial fidelity of reconstructed signals [33, 34]. Second, existing methods provide insufficient time-frequency feature preservation. EEG signals exhibit rich multi-scale spectral characteristics within the physiologically relevant 4–40 Hz range, with different frequency bands (theta: 4–8 Hz, alpha: 8–13 Hz, beta: 13–30 Hz, low-gamma: 30–40 Hz) reflecting distinct neural processes [35]. Motor imagery particularly involves modulation of mu (8–13 Hz) and beta (13–30 Hz) rhythms, known as event-related desynchronization (ERD) and synchronization (ERS) [36]. Existing architectures use fixed frequency band divisions or single-scale feature extraction, unable to adaptively emphasize task-relevant frequency bands, leading to the loss of critical spectral features [37, 38]. Third, task conditioning remains inadequately utilized. Existing EEG diffusion methods exhibit deficiencies in handling task conditioning: some methods completely omit task conditions (e.g., BrainDiff [26]), resulting in generated samples that lack task-discriminative structure and provide limited utility for motor imagery classification augmentation; others introduce task conditions but only as simple embedding concatenations, failing to fully exploit their potential for guiding the generation of class-discriminative samples. As a key technique in data augmentation, task conditioning should be able to guide the model to generate class-relevant samples containing richer discriminative information, thereby more effectively supplementing training data [39, 40, 41].

To overcome these shortcomings, this study proposes DTTD, a Dynamic Topology-aware and Task-synergistic Diffusion Model for conditional EEG data augmentation. DTTD achieves improvements through four key innovations. First, we introduce task-conditioned dynamic topology learning. A learnable Graph Neural Network (GNN) [42, 43] discovers spatial dependencies among electrodes in an end-to-end manner, while task embeddings modulate the learned adjacency matrix via multi-head attention, enabling adaptation to task-specific functional connectivity. Second, we design multi-scale frequency decoupling with task-adaptive weights. Leveraging neurophysiological knowledge, EEG is decomposed into multiple frequency bands (theta, alpha, low-beta, high-beta, low-gamma) via FFT, each processed by dedicated CNN encoders. A task-aware weighting network dynamically emphasizes relevant bands—e.g., alpha/beta for motor imagery—enabling adaptive multi-scale time-frequency modeling. Third, we employ gated spatio-temporal-spectral fusion. A gating mechanism replaces naive concatenation to merge topology-enhanced and frequency-enhanced features, allowing context-dependent emphasis on spatial or spectral aspects. Fourth, we develop efficient light-noise training with direct X_0 prediction. Training with controlled light noise ($\eta = 0.2$, 20% noise scale) preserves strong input-target correlation, and the model directly predicts clean signals (X_0), enabling fast single-step inference with throughput of approximately 71 samples/second.

Extensive experiments on three datasets demonstrate the effectiveness of DTTD. On BCI Competition IV 2a, DTTD augmentation improves within-subject classification accuracy to 78.38% (+3.90 pp over the 22-channel refer-

ence, $p < 0.01$) and cross-session accuracy to 65.70% (+2.62 pp, $p < 0.01$). On HGD, DTTD exceeds the 128-channel reference by 6.08 pp in cross-subject and 3.53 pp in cross-session scenarios. On PhysioNet MI, DTTD achieves comparable performance to the 64-channel reference in cross-subject evaluation. Ablation studies confirm that multi-scale frequency decoupling is the largest contributor (+7.37 pp on BCI2a, +3.48 pp on HGD), followed by dynamic topology learning (+4.18 pp on BCI2a, +2.99 pp on HGD).

The remainder of this paper is organized as follows. Section 2 describes the problem formulation and the DTTD architecture in detail. Section 3 presents the experimental setup and results including reconstruction quality evaluation, classification augmentation, and ablation studies. Section 4 discusses the findings and their implications. Section 5 concludes the paper.

2. Materials and Methods

2.1. Problem Formulation

Given an EEG dataset $\mathcal{D} = \{(X_i^{\text{in}}, X_i^{\text{full}}, y_i)\}_{i=1}^N$, where:

- $X_i^{\text{in}} \in \mathbb{R}^{C_{\text{in}} \times T}$ denotes the input low-density channel signals (C_{in} channels, T time points),
- $X_i^{\text{full}} \in \mathbb{R}^{C_{\text{full}} \times T}$ represents the complete high-density channel data (C_{full} channels, $C_{\text{full}} > C_{\text{in}}$),
- $y_i \in \{1, 2, \dots, K\}$ is the motor imagery task label (K classes).

The objective is to learn a conditional generative model:

$$p_{\theta}(X^{\text{full}}|X^{\text{in}}, y) \quad (1)$$

that generates complete high-density EEG signals from sparse-channel inputs and task labels. For example, on the BCI Competition IV 2a dataset, $C_{\text{in}} = 9$ and $C_{\text{full}} = 22$; on the High Gamma Dataset, $C_{\text{in}} = 16$ and $C_{\text{full}} = 128$; on PhysioNet MI, $C_{\text{in}} = 16$ and $C_{\text{full}} = 64$. The core value of this method lies in: by introducing task-discriminative information, the generated samples can effectively supplement the diversity of training data, thereby improving the generalization capability of classifiers across different evaluation scenarios.

2.2. DTTD Architecture Overview

Figure 1 illustrates the overall architecture of DTTD, comprising a five-stage data flow: (a) Raw EEG—paired sparse-channel and full-channel signals as input; (b) Forward Diffusion—injects Gaussian noise into X_0 over $T=1000$ steps per schedule, converting it to pure noise for reverse learning; (c) Backbone—integrates conditional DDPM with topology/frequency modules to ensure generated samples match raw data’s spatio-temporal features; (d) Reverse Denoising—using X_0 and 20% noisy data, the trained backbone with task-conditional guidance progressively removes noise to accurately generate spatial and time-frequency features; (e) Generated EEG—outputs the augmented high-density EEG signals.

Within the backbone (c), the DTTD framework integrates four core modules: (1) a dynamic topology-aware module with task modulation (upper branch of Fig. 1(c)), which learns adaptive electrode adjacency matrices and applies graph convolution to capture task-specific spatial dependencies; (2) a frequency decoupling module with task-adaptive band weighting (lower branch of Fig. 1(c)), which independently processes different frequency bands within the 4–40 Hz range while preserving temporal information; (3) a gated feature fusion mechanism (junction in Fig. 1(c)), which dynamically combines topology and frequency features, adaptively balancing contributions from both modalities; (4) an enhanced U-Net denoising network (main structure of Fig. 1(c1)), which serves as the backbone for signal generation, fusing time-step and task-condition embeddings at each residual block. Unlike standard diffusion models, DTTD employs a controlled light-noise training strategy with direct X_0 prediction, enabling efficient single-step or few-step inference.

2.3. Dynamic Topology-aware Module with Task Modulation

Traditional EEG analysis assumes fixed spatial relationships based on physical electrode distances or predefined functional connectivity templates. However, neural functional connectivity is inherently dynamic, reorganizing according to cognitive states and task demands [44]. To capture this phenomenon, we propose a learnable topology modeling module that adapts to data-driven spatial dependencies and incorporates task-specific modulation.

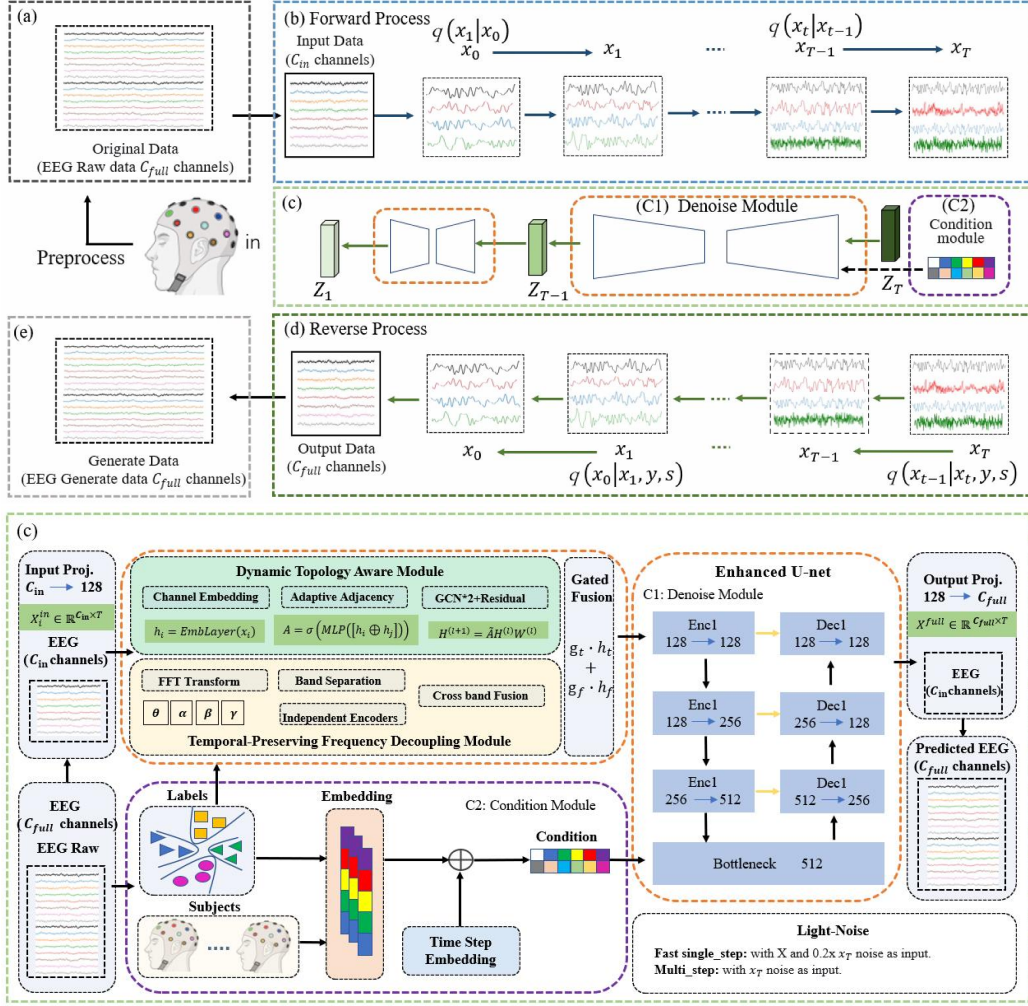


Figure 1: Overall architecture of DTTD. (a) Raw EEG. (b) Forward Diffusion: Injects Gaussian noise into X_0 over $T=1000$ steps per schedule, converting it to pure noise for learning. (c) Backbone: Integrates conditional DDPM with topology/frequency modules to ensure generated samples match raw data’s spatio-temporal features. (d) Reverse Denoising: Using X_0 and 20% noisy data, the trained backbone with task-conditional guidance progressively removes noise to accurately generate spatial and time-frequency features. (e) Generated EEG.

2.3.1. Channel Embeddings with Spatial Positional Encoding

For input EEG $X \in \mathbb{R}^{C \times T}$, we first extract channel-wise feature embeddings through a linear projection that maps each channel’s temporal features to a high-dimensional embedding $h_i \in \mathbb{R}^d$. To preserve spatial information, positional encodings based on the 10-20 electrode system coordinates are added to the channel embeddings, where the positional encoding for electrode i is derived from its 3D coordinate pos_i in the 10-20 system.

A key innovation is the task modulation mechanism. Given task embedding $e_y \in \mathbb{R}^{d_{task}}$ from a learnable task encoder, a two-layer MLP projects the task embedding to the channel embedding space and adds it to the spatially-encoded features. This modulation enables the topology module to learn task-specific connectivity patterns—for example, emphasizing motor cortex connections during motor imagery tasks.

2.3.2. Adaptive Adjacency Matrix Construction

To model pairwise channel interactions, we concatenate task-modulated embedding pairs and pass them through a multi-layer perceptron (MLP) with sigmoid activation, producing a scalar $A_{ij} \in [0, 1]$ that represents the learned connection strength between channels i and j . The adjacency matrix is symmetrized by averaging with its transpose.

This soft, continuous adjacency matrix is dynamically computed for each input sample conditioned on the task label.

2.3.3. Normalized Graph Convolution

To aggregate information from spatially related channels, we perform graph convolution using the learned adjacency matrix. First, we compute the degree-normalized adjacency matrix:

$$\tilde{A} = D^{-1}A \quad (2)$$

where D is the diagonal degree matrix. We then apply normalized graph convolution to the input features:

$$X_{\text{topo}} = \tilde{A}XW \quad (3)$$

Multiple graph convolution layers are stacked to capture multi-hop spatial dependencies, with residual connections and layer normalization to enhance expressiveness.

2.4. Temporal-Preserving Frequency Decoupling Module

EEG signals contain multiple physiologically meaningful frequency bands, each associated with distinct neural processes. Theta (4–8 Hz) relates to drowsiness and memory, alpha (8–13 Hz) to relaxed wakefulness, low-beta (13–20 Hz) and high-beta (20–30 Hz) to active thinking and motor control, and low-gamma (30–40 Hz) to cognitive processing. Motor imagery specifically involves modulation of mu and beta rhythms.

We use FFT to decompose EEG signals into multiple frequency bands. To avoid spectral leakage, a Hanning window is applied before FFT. For each band b , we define smooth Gaussian masks M_b (rather than binary masks) to avoid Gibbs phenomena and boundary artifacts. The frequency bands are defined within the 4–40 Hz range, consistent with the preprocessing bandpass filter. The band decomposition can be formalized as:

$$X_b = \text{IFFT}(\text{FFT}(X) \odot M_b), \quad b \in \{\text{theta, alpha, low-beta, high-beta, low-gamma}\} \quad (4)$$

Dedicated convolutional encoders are applied to each frequency band, with all convolutions using padding to preserve temporal length. An attention mechanism conditioned on the task label is then employed to learn band importance: global average pooling is first applied to extract band-level features, which are then passed through a band attention module with softmax normalization to produce attention weights α .

A key innovation is the task-conditioned attention modulation: a linear projection maps the task embedding e_y to a band importance bias, which is added to the attention logits before softmax normalization, producing modulated weights α^{mod} . This allows the model to dynamically emphasize task-relevant frequency bands—for example, alpha/beta bands during motor imagery.

The fused frequency features are computed as an attention-weighted sum, with a residual connection using learnable scaling to ensure stable training.

2.5. Gated Spatio-Temporal-Spectral Feature Fusion

To effectively combine topology-enhanced features X_{topo} and frequency-enhanced features X_{freq} , we design a gated fusion module. The topology and frequency features are first concatenated along the channel dimension and passed through a convolution layer with sigmoid activation to produce a gating map $g \in [0, 1]$. The fused output is then computed as a convex combination: $X_{\text{fused}} = g \odot X_{\text{topo}} + (1 - g) \odot X_{\text{freq}}$, where $\odot$ denotes element-wise multiplication. The gating mechanism enables the model to adaptively emphasize topology or frequency features based on input content.

2.6. Enhanced U-Net Backbone

The denoising network employs an enhanced U-Net architecture that supports both time and task conditioning at every layer.

2.6.1. Architecture Overview

The U-Net consists of an encoder path, a bottleneck, and a decoder path with skip connections. The encoder comprises three residual blocks with progressively increasing channel dimensions ($d \rightarrow 2d \rightarrow 4d$), each followed by max pooling for downsampling. The bottleneck is a residual block at the lowest resolution that captures global context. The decoder comprises three residual blocks with skip connections from the encoder, using linear interpolation for upsampling. The architecture incorporates dual conditioning through embedding vectors: time embedding uses Transformer-style positional encoding for the diffusion timestep, and task embedding uses a learnable embedding layer that maps task labels to dense vectors. Both conditions are injected into the network via adaptive layer normalization (AdaLN).

2.6.2. Conditioned Residual Blocks

Each residual block applies the following operations sequentially: layer normalization of the input, followed by a convolution; addition of the projected time embedding e_t and task embedding e_y via linear projections; SiLU activation; a second layer normalization; and finally a second convolution with a residual connection. This dual-conditioning design allows the network to dynamically adapt its behavior based on both the diffusion timestep and the target task.

2.7. Controlled Light-Noise Training with Direct X_0 Prediction

Standard diffusion models require hundreds of denoising steps during inference, limiting practical applicability. We propose a controlled light-noise training strategy specifically designed for channel reconstruction tasks where partial information is always available.

2.7.1. Controlled Light-Noise Forward Process

Unlike standard DDPM that fully corrupts the signal over multiple steps, we add only controlled light noise to the input channels during training:

$$X_{\text{noisy}} = X_{\text{in}} + \eta \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \quad (5)$$

where $\eta = 0.2$ (20% noise scale). This strategy maintains strong input-output correlation while providing regularization against overfitting.

2.7.2. Direct X_0 Prediction

Rather than predicting the noise ϵ , the model directly predicts the clean 22-channel signal conditioned on the task label:

$$\hat{X}_0 = f_{\theta}(X_{\text{noisy}}, t, y) \quad (6)$$

The reconstruction loss is:

$$\mathcal{L}_{\text{recon}} = \|X_0 - \hat{X}_0\|_2^2 \quad (7)$$

This formulation supports efficient single-step inference: given 9-channel input and a task label, the model directly outputs the 22-channel generated signal without iterative denoising.

2.7.3. Auxiliary Classification Regularization

To encourage task-discriminative feature generation, we incorporate an auxiliary classification head C_{ϕ} that provides additional supervision during training. The auxiliary classifier is frozen after initialization to prevent overfitting to generated data:

$$\mathcal{L}_{\text{cls}} = \text{CrossEntropy}(C_{\phi}(\hat{X}_0), y) \quad (8)$$

The total loss with progressive classification regularization is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{recon}} + \lambda_{\text{cls}}(e) \cdot \mathcal{L}_{\text{cls}} \quad (9)$$

where $\lambda_{\text{cls}}(e) = \lambda_{\text{target}} \cdot \min(1, e/e_{\text{warmup}})$ gradually increases from 0 to λ_{target} over warmup epochs to prevent early training instability. The loss supports multiple reconstruction objectives: L2 loss (default, $\|X_0 - \hat{X}_0\|_2^2$) for smooth reconstructions, L1 loss ($\|X_0 - \hat{X}_0\|_1$) for sharper reconstructions, and Huber loss (smooth L1) for robustness to outliers. Key hyperparameters: $\lambda_{\text{cls}} = 2.0$ (target), warmup epochs = 30, classification loss clipping at 3.0 to prevent training instability.

2.8. Sampling with Task Conditioning

During inference, DTTD generates task-specific augmented data conditioned on task labels. Given input X^{in} and task label y , the model predicts the full-channel output:

$$\hat{X}^{\text{full}} = f_{\theta}(X^{\text{in}} + \eta \cdot \epsilon, t = 0, y) \quad (10)$$

This enables efficient generation of task-discriminative augmented data for downstream classification training. During the training phase, DTTD generates high-density EEG samples using known labels as conditions, and these generated samples are combined with the original high-density data to form the augmented training set. The algorithm supports both fast single-step inference (recommended for real-time applications) and optional multi-step DDIM sampling (for higher quality). After augmented data generation, the generated samples are merged with the original training data for classifier training.

Algorithm 1 DTTD Sampling for EEG Data Augmentation

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1: Input: Input  $X^{\text{in}}$ , task label  $y$ , num_steps  $N$ 
2: Output: Reconstructed EEG  $\hat{X}^{\text{full}}$ 
3: // Fast path: single-step inference
4: if  $N = 1$  then
5:    $\hat{X}^{\text{full}} \leftarrow f_{\theta}(X^{\text{in}} + \eta \cdot \epsilon, t = 0, y)$ 
6:   return  $\hat{X}^{\text{full}}$ 
7: end if
8: // Multi-step path: DDIM sampling
9:  $X_T \sim \mathcal{N}(0, I)$  // Start from pure noise
10: for  $t = T, T - 1, \dots, 1$  do
11:   Predict  $X_0$ :  $\hat{X}_0 \leftarrow f_{\theta}(X_t^{\text{in}}, t, y)$ 
12:   Interpolate:  $X_{t-1} \leftarrow \alpha_t \hat{X}_0 + \beta_t X_t^{\text{in}}$ 
13: end for
14: return  $\hat{X}_0$ 

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3. Experiments and Results

3.1. Datasets and Preprocessing

3.1.1. BCI Competition IV Dataset 2a

The primary evaluation dataset is BCI Competition IV Dataset 2a [45], a widely-used benchmark for motor imagery BCIs. It contains EEG recordings from 9 healthy subjects performing four motor imagery tasks: left hand, right hand, both feet, and tongue. Data were acquired using 22 Ag/AgCl electrodes positioned according to the international 10-20 system at a sampling rate of 250 Hz. Each subject completed 288 trials (72 per class) across two sessions recorded on different days.

For the EEG data augmentation task, we select 9 motor cortex channels as input: C3, Cz, C4, FC3, FCz, FC4, CP3, CPz, CP4. These channels cover the sensorimotor cortex bilaterally and represent a typical low-density electrode configuration. The goal is to generate complete 22-channel EEG samples from these 9 sparse input channels.

3.1.2. PhysioNet MI Dataset

For further validation of DTTD’s effectiveness, experiments are conducted on the PhysioNet Motor Movement/Imagery dataset [46]. This dataset contains 64-channel EEG recordings from 109 subjects performing four motor imagery tasks: left hand, right hand, both fists, and both feet, at 160 Hz. Each subject completed 6 motor imagery runs (approximately 90 trials in total, ~ 22 per class). We select 16 channels covering the motor cortex as input to generate augmented 64-channel data.

3.1.3. High Gamma Dataset (HGD)

The HGD [38] contains 128-channel EEG recordings (plus 5 non-EEG channels) from 14 healthy subjects performing four motor imagery tasks: right hand, left hand, rest, and feet, at 500 Hz. Each subject completed approximately 1,000 4-second trials across 13 runs. The training set comprises the first 11 runs (~ 880 trials per subject) and the test set uses the remaining 2 runs (~ 160 trials per subject). Due to memory constraints, the raw data is resampled to 250 Hz. We select 16 channels covering the standard 10-20 motor cortex montage as the low-density input, with the full 128-channel EEG data as the reference.

3.1.4. Data Preprocessing

EEG signals undergo standardized preprocessing using the MNE-Python library [16]. First, a 4th-order Butterworth bandpass filter (4–40 Hz) removes low-frequency drifts and high-frequency noise [47]. Second, common average reference (CAR) spatial filtering reduces common-mode artifacts. Third, trials exceeding $\pm 100 \mu\text{V}$ are excluded as artifact rejection. Fourth, channel-wise exponential moving standardization with smoothing parameter $\alpha = 0.999$ stabilizes amplitude distributions. Finally, for BCI2a, we extract 4-second epochs from 2–6 seconds post-cue, yielding samples of shape (22, 1000); for HGD, 4-second epochs resampled to 250 Hz yield samples of shape (128, 1000); for PhysioNet MI, 4-second epochs yield samples of shape (64, 640).

3.2. Experimental Setup

3.2.1. Implementation Details

DTTD is implemented in PyTorch and trained on an NVIDIA RTX 3090 GPU. The model architecture uses input channels $C_{\text{in}} = 9$, output channels $C_{\text{out}} = 22$, embedding dimension $d = 128$, task dimension $d_{\text{task}} = 128$, 8 attention heads, dropout = 0.1. The diffusion schedule employs $T = 1000$ timesteps with a linear schedule ($\beta_{\text{start}} = 10^{-4}$, $\beta_{\text{end}} = 0.02$). The U-Net backbone consists of 3 encoder blocks and 3 decoder blocks with skip connections, hidden dimensions [128, 256, 512], and GroupNorm with 8 groups. The topology module uses embedding dimension $d = 128$, 2 graph convolution layers, 8-head attention for adjacency learning, and task modulation via a 2-layer MLP. The frequency attention module decomposes into 5 frequency bands (theta: 4–8Hz, alpha: 8–13Hz, low-beta: 13–20Hz, high-beta: 20–30Hz, low-gamma: 30–40Hz), with temporal-preserving band convolutions, attention-based fusion with task modulation, and learnable residual scaling ($\gamma = 0.5$). Training uses a controlled light-noise strategy with $\eta = 0.2$, batch size 32, learning rate 2×10^{-4} , AdamW optimizer with weight decay 10^{-5} , and 200 epochs.

3.2.2. Evaluation Protocols

We employ multiple evaluation scenarios with strict data partitioning to ensure training/testing independence. Different datasets adopt different combinations of evaluation scenarios according to their data structure characteristics.

For BCI Competition IV 2a (9-ch $\rightarrow$ 22-ch, 9 subjects, 2 sessions), three evaluation scenarios are adopted. In the within-subject scenario, both sessions are combined for each subject and split via 5-fold stratified cross-validation, with 80% of trials for training and 20% for testing in each fold. In the cross-session scenario, session T is used exclusively for training and session E exclusively for testing, ensuring complete temporal separation. In the cross-subject scenario, leave-one-subject-out (LOSO) cross-validation is performed across all 9 subjects, where 8 subjects’ data are used for training and the remaining subject’s data for testing in each fold.

For the High Gamma Dataset (HGD) (16-ch $\rightarrow$ 128-ch, 14 subjects, 13 runs), two scenarios are used. In the cross-session scenario, the first 11 runs (~ 880 trials per subject) serve as training data and the remaining 2 runs (~ 160 trials per subject) as test data. In the cross-subject scenario, LOSO cross-validation is performed across all 14 subjects.

For PhysioNet MI (16-ch $\rightarrow$ 64-ch, 109 subjects), only the cross-subject scenario is evaluated using 10-fold cross-subject cross-validation, ensuring no subject’s data appears in both training and testing sets.

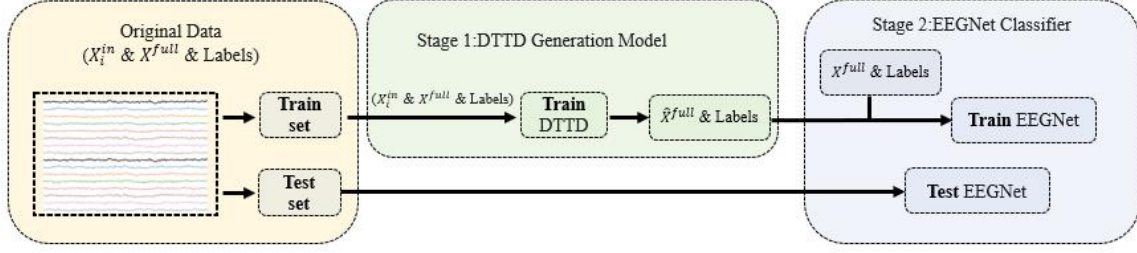


Figure 2: Two-stage data augmentation pipeline. Stage 1: DTTD training and generation on training data only. Stage 2: EEGNet classifier training on augmented data and evaluation on held-out test data.

For each scenario, an EEGNet classifier [48] is trained independently on the original sparse-channel data, the augmented high-density channel data, or the original high-density channel data (without augmentation). All classifiers are trained from scratch with the same hyperparameters.

Each evaluation scenario follows a unified two-stage data augmentation pipeline with strict separation between generative model and classifier data usage, as illustrated in Figure 2. In Stage 1, DTTD is trained on the training set’s paired (sparse-channel, full-channel) data with task labels, and then the trained DTTD generates full-channel EEG samples conditioned on the training set’s sparse-channel inputs and known task labels, using only training set data. In Stage 2, the generated full-channel samples are combined with the original high-density training data to form the augmented training set for EEGNet classifier training, and the classifier is evaluated on real full-channel test data that is never used in DTTD training or generation. The key principle is: DTTD is trained and generates only on the training set; the classifier is trained only on the augmented training set; neither uses any information from the test set.

For statistical analysis, reconstruction quality evaluation uses the Wilcoxon signed-rank test for significance between DTTD and other methods, and classification performance evaluation uses paired t-tests for comparing DTTD augmentation with reference channel configurations.

3.2.3. Baseline Methods

We compare against representative reconstruction and augmentation approaches. Spherical spline interpolation [12] is a traditional signal processing method assuming smooth potential distributions. Kriging [14] is a geostatistical interpolation method using spatial covariance models. Conditional VAE (CVAE) is a conditional variational autoencoder with KL annealing. TimeGAN is a time-series GAN with supervised and unsupervised losses [49]. EEGDiff [24] is a diffusion-based EEG generation model with static electrode adjacency. BrainDiff [26] is a diffusion model with frequency-aware processing but no task conditioning.

3.3. Reconstruction Quality Evaluation

We evaluate reconstruction quality metrics on 200 randomly selected samples from each of the three datasets. Wilcoxon signed-rank test is used for statistical significance between DTTD and other methods.

DTTD achieves optimal reconstruction quality in large-scale channel expansion scenarios. On HGD (16→128 channels, 8× expansion), DTTD topology similarity reaches 0.967, significantly outperforming EEGDiff (0.930), BrainDiff (0.935), CVAE (0.785), cGAN (0.775), Kriging (0.462), and Spline (0.462) (Wilcoxon test, $p < 0.001$), demonstrating DTTD’s ability to accurately reconstruct spatial topology structures even in large-scale channel expansion scenarios. On PhysioNet MI (16→64 channels), DTTD topology similarity is 0.949, substantially outperforming CVAE (0.791), cGAN (0.762), Kriging (0.496), and Spline (0.318). On BCI2a (9→22 channels), DTTD topology similarity is 0.910, significantly outperforming cGAN (0.899), CVAE (0.892), Kriging (0.616), and Spline (0.457).

Regarding the comparative analysis of dynamic topology and task conditioning, EEGDiff (static electrode adjacency) and BrainDiff (frequency-aware without task conditioning) serve as comparison models for DTTD, both outperforming non-diffusion methods such as CVAE/cGAN across all three datasets, validating the effectiveness of the diffusion framework. On HGD (16→128 channels), DTTD improves topology similarity by 3.7 percentage points over EEGDiff (0.967 vs 0.930) and PSD correlation by 1.3 percentage points (0.983 vs 0.970), demonstrating the

Table 1: Reconstruction Quality Metrics Across Three Datasets (200 samples each)

Dataset	Method	Topo. Sim.↑	NMSE↓	PSD Corr.↑	Freq. Sim.↑
BCI2a (9→22)	DTTD	0.910	0.131	0.947	0.978
BCI2a	EEGDiff	0.901	0.171	0.925	0.912
BCI2a	BrainDiff	0.908	0.163	0.915	0.902
BCI2a	cGAN	0.899	0.508	0.801	0.367
BCI2a	CVAE	0.892	0.486	0.618	0.415
BCI2a	Kriging	0.616	0.084	0.976	0.960
BCI2a	Spline	0.457	3.506	0.939	0.899
HGD (16→128)	DTTD	0.967	0.051	0.983	0.917
HGD	EEGDiff	0.930	0.082	0.970	0.892
HGD	BrainDiff	0.935	0.101	0.972	0.895
HGD	CVAE	0.785	0.608	0.686	0.433
HGD	cGAN	0.775	0.610	0.619	0.368
HGD	Kriging	0.462	0.779	0.916	0.763
HGD	Spline	0.462	0.779	0.916	0.763
PhysioNet (16→64)	DTTD	0.949	0.312	0.918	0.824
PhysioNet	BrainDiff	0.933	0.379	0.904	0.727
PhysioNet	EEGDiff	0.931	0.392	0.901	0.722
PhysioNet	CVAE	0.791	0.876	0.750	0.521
PhysioNet	cGAN	0.762	0.878	0.733	0.452
PhysioNet	Kriging	0.496	0.932	0.799	0.641
PhysioNet	Spline	0.318	0.957	0.821	0.686

BCI2a: Wilcoxon test DTTD vs CVAE/cGAN/interpolation $p < 0.001$;

HGD: DTTD vs all baselines $p < 0.001$; PhysioNet: DTTD vs CVAE/cGAN/interpolation $p < 0.001$

critical role of dynamic topology learning in large-scale channel expansion. In smaller expansion scenarios such as BCI2a (9→22 channels) and PhysioNet MI (16→64 channels), the gap between EEGDiff/BrainDiff and DTTD is smaller (<4 percentage points), indicating that the advantage of dynamic topology becomes more pronounced with increasing channel expansion ratio.

DTTD also performs best in spectral feature preservation. DTTD achieves the highest PSD correlation (0.983) and frequency similarity (0.917) on HGD, and similarly leads non-diffusion methods on BCI2a (PSD correlation 0.947, frequency similarity 0.978). Although CVAE/cGAN show relatively high topology similarity (~0.90), their PSD correlation (0.618–0.801) and frequency similarity (0.367–0.415) are significantly lower than DTTD, indicating they learn “average” spatial patterns rather than sample-specific reconstructions.

In contrast, traditional interpolation methods exhibit notable limitations. Kriging achieves high PSD correlation (0.976) on BCI2a but poor topology similarity (0.616), and performance degrades sharply in large-scale channel expansion scenarios (HGD, PhysioNet MI). Spline interpolation exhibits $\text{NMSE} > 0.7$ on all datasets due to numerical instability from extrapolation—when input channels cannot cover the spatial range of output channels, cubic polynomial extrapolation causes values to grow dramatically.

3.4. Motor Imagery Classification Performance

3.4.1. Within-Subject Classification

Using EEGNet in the within-subject scenario, the original 22-channel data accuracy is $74.48 \pm 5.90\%$. When the training set is augmented with DTTD-generated 22-channel data combined with original high-density data (using true labels for conditional generation), the classification accuracy reaches $78.38 \pm 8.60\%$, a 3.90 percentage point improvement over the 22-channel reference ($p < 0.01$).

DTTD augmentation at $78.38 \pm 8.60\%$ exceeds the 22-channel reference ($74.48 \pm 5.90\%$) by 3.90 pp, performing

Table 2: BCI2a Within-Subject Classification Accuracy (%) per Subject

Method	S1	S2	S3	S4	S5	S6	S7	S8	S9	Mean	κ
22-ch Ref.	69.98	65.27	80.38	72.05	81.09	65.97	77.94	77.96	79.68	74.48±5.90	0.66
Spline Aug.	76.73	59.71	85.59	61.80	73.26	61.12	75.70	79.34	81.94	72.80±9.09	0.64
Kriging Aug.	80.90	59.55	88.54	69.97	74.83	62.51	73.77	82.82	86.80	75.52±9.65	0.67
CVAE Aug.	68.50	63.20	78.90	70.50	79.80	64.80	76.20	76.80	77.50	73.15±6.28	0.62
cGAN Aug.	73.79	65.45	77.09	71.53	80.04	68.24	75.34	72.39	86.28	74.46±5.89	0.66
EEGDiff Aug.	81.06	63.19	85.77	72.23	79.52	67.54	75.86	81.08	89.23	77.28±7.95	0.68
BrainDiff Aug.	80.03	64.58	88.02	72.75	79.00	66.15	77.24	82.98	88.72	77.72±8.12	0.68
DTTD Aug.	81.95	65.27	89.75	70.50	81.09	67.88	80.02	79.87	90.45	78.38±8.60	0.71
vs 22-ch	+11.97	-0.87	+9.37	-1.55	-0.53	+1.91	+2.08	+1.91	+10.77	+3.90	-

Table 3: BCI2a Cross-Session Classification Accuracy (%) per Subject

Method	S1	S2	S3	S4	S5	S6	S7	S8	S9	Mean	κ
22-ch Ref.	72.22	44.10	77.78	59.38	38.89	44.44	71.53	78.13	81.25	63.08±15.75	0.51
Spline Aug.	60.76	45.83	70.49	52.08	59.03	52.08	66.67	56.60	63.89	58.60±7.39	0.45
Kriging Aug.	67.01	54.86	71.53	53.12	61.46	56.60	62.85	65.97	68.40	62.42±6.06	0.50
CVAE Aug.	69.44	52.08	69.79	59.72	69.44	55.56	66.32	68.75	70.83	64.66±6.63	0.53
cGAN Aug.	60.07	56.94	73.61	56.94	70.83	54.86	68.75	65.28	61.81	63.23±6.34	0.51
EEGDiff Aug.	73.91	51.34	78.77	49.25	69.39	55.85	73.21	66.27	72.17	65.57±10.13	0.55
BrainDiff Aug.	70.14	50.35	81.25	64.24	68.75	57.64	68.75	69.44	66.32	65.32±8.12	0.55
DTTD Aug.	69.10	53.82	82.99	60.76	70.14	57.29	69.44	63.54	64.24	65.70±8.08	0.57
vs 22-ch	-3.12	+9.72	+5.21	+1.38	+31.25	+12.85	-2.09	-14.59	-17.01	+2.62	-

Table 4: HGD Cross-Session Classification Accuracy (%) per Subject

Method	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	Mean	κ
128-ch	66.25	66.25	78.13	88.75	86.25	83.75	83.75	85.63	95.00	88.75	77.50	82.50	90.57	93.75	83.34±8.52	0.78
Spline	63.75	68.13	86.88	90.63	81.88	78.13	75.00	83.75	88.75	83.13	76.25	81.25	89.31	68.13	79.64±8.19	0.73
Kriging	70.00	71.88	86.25	90.63	93.13	70.63	78.13	86.25	90.00	85.63	74.38	91.88	86.16	65.00	81.42±9.10	0.75
CVAE	73.75	76.88	72.50	87.50	61.88	75.63	72.50	45.00	93.75	88.13	70.00	66.25	84.91	64.38	73.79±12.14	0.65
cGAN	51.88	71.88	83.75	86.25	78.75	72.50	60.00	65.00	80.63	81.88	41.25	70.00	81.76	46.88	69.46±14.00	0.59
EEGDiff	81.25	81.25	90.00	89.38	92.50	80.63	86.25	78.75	96.88	86.88	76.25	91.25	90.57	60.63	84.46±8.74	0.79
BrainDiff	76.88	73.75	92.50	95.63	95.00	78.13	86.88	83.75	95.63	90.00	78.75	93.13	91.82	70.63	85.89±8.45	0.81
DTTD	86.88	85.21	95.86	91.38	93.75	90.00	89.13	84.38	97.25	89.50	83.95	93.63	93.57	85.23	86.87±7.56	0.82
vs 128ch	+20.63	+18.96	+17.73	+2.63	+7.50	+6.25	+5.38	-1.25	+2.25	+0.75	+6.45	+11.13	+3.00	-8.52	+3.53	-

best among all augmentation methods, confirming the effectiveness of task-conditioned generation for classification augmentation.

3.4.2. Cross-Session Classification

In the cross-session scenario, DTTD augmentation shows positive gains across both datasets. On BCI2a (train on session T, test on session E), the 22-channel reference accuracy is 63.08±15.75%, and DTTD augmentation improves it to 65.70±8.08% (+2.62 pp, $p < 0.01$), indicating that the model learns enhanced session-invariant representations through topology-aware reconstruction. On HGD, the 128-channel reference already reaches 83.34±8.52% in the high-accuracy regime, and DTTD augmentation achieves 86.87±7.56%, exceeding the 128-channel reference by 3.53 pp.

Among 9 subjects on BCI2a, 5 exceed the 22-channel baseline after DTTD augmentation, with S5 showing the most significant improvement (+31.25 pp), whose inter-session variability is large (22-channel baseline only 38.89%). S1 (-3.12 pp), S7 (-2.09 pp), S8 (-14.59 pp), and S9 (-17.01 pp) show lower accuracy after augmentation, where S8 and S9 already have high 22-channel baselines (78.13% and 81.25%), leaving limited room for improvement.

Table 4 presents per-subject cross-session results on HGD. In the high-accuracy regime, differences among meth-

Table 5: BCI2a Cross-Subject Classification Accuracy (%) per Subject

Method	S1	S2	S3	S4	S5	S6	S7	S8	S9	Mean	κ
22-ch Ref.	61.46	53.13	66.15	50.52	53.47	51.91	69.44	64.76	56.42	58.58 $\pm$ 6.60	0.45
Spline Aug.	61.46	55.03	68.23	43.75	57.64	54.69	68.58	68.06	56.94	59.38 $\pm$ 7.72	0.46
Kriging Aug.	61.98	48.09	68.58	44.44	52.95	51.91	69.44	64.06	57.29	57.64 $\pm$ 8.43	0.44
CVAE Aug.	62.85	52.78	56.94	51.74	57.12	55.90	70.31	67.19	55.56	58.93 $\pm$ 6.06	0.45
cGAN Aug.	60.42	47.57	64.93	48.09	64.41	53.47	63.89	63.54	53.99	57.81 $\pm$ 6.70	0.44
EEGDiff Aug.	59.20	50.35	67.36	46.70	57.29	53.47	70.49	61.81	56.94	58.18 $\pm$ 7.21	0.44
BrainDiff Aug.	59.07	53.86	72.26	47.78	55.94	51.95	69.66	66.36	56.29	59.24 $\pm$ 7.90	0.46
DTTD Aug.	60.94	51.91	68.40	47.40	54.17	52.78	72.40	67.19	62.85	59.78$\pm$8.13	0.46
vs 22-ch	-0.52	-1.22	+2.25	-3.12	+0.70	+0.87	+2.96	+2.43	+6.43	+1.20	-

Table 6: HGD Cross-Subject Classification Accuracy (%) per Subject

Method	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	Mean	κ
128-ch	81.25	54.99	46.15	63.95	74.74	62.89	72.02	54.55	58.85	42.96	52.40	53.62	49.64	73.37	60.10 $\pm$ 11.21	0.47
Spline	75.62	54.89	47.50	47.25	80.66	55.33	64.68	66.71	64.81	41.11	30.87	67.02	46.72	72.40	58.25 $\pm$ 13.81	0.44
Kriging	68.96	38.21	35.19	65.84	61.09	55.23	71.53	66.22	63.27	35.56	59.90	73.87	40.25	45.19	55.74 $\pm$ 13.51	0.41
CVAE	79.17	67.66	57.69	51.04	79.98	51.65	64.88	75.18	62.98	56.32	49.33	68.18	33.06	51.54	60.62 $\pm$ 12.62	0.48
cGAN	48.12	46.34	33.17	70.21	77.59	46.41	69.74	66.09	57.31	62.28	37.60	75.22	55.68	71.54	58.38 $\pm$ 13.73	0.45
EEGDiff	84.58	69.52	58.17	50.47	81.91	48.64	65.58	60.93	59.62	64.44	58.65	63.36	57.87	74.13	64.13 $\pm$ 10.10	0.52
BrainDiff	75.21	52.42	43.17	60.91	72.24	54.26	70.14	61.06	67.02	63.82	47.50	55.74	35.56	76.25	59.66 $\pm$ 11.83	0.46
DTTD	94.17	60.97	54.04	67.65	87.14	62.02	59.72	68.43	65.10	68.76	39.90	73.48	48.18	76.92	66.18$\pm$13.77	0.55
vs 128ch	+12.92	+5.98	+7.89	+3.70	+12.40	-0.87	-12.30	+13.88	+6.25	+25.80	-12.50	+19.86	-1.46	+3.55	+6.08	-

ods are limited. Among traditional interpolation methods, Kriging augmentation (81.42 $\pm$ 9.10%) outperforms Spline augmentation (79.64 $\pm$ 8.19%), but both remain below the 128-channel reference (83.34 $\pm$ 8.52%). Among deep generative methods, BrainDiff augmentation (85.89 $\pm$ 8.45%) and EEGDiff augmentation (84.46 $\pm$ 8.74%) approach the 128-channel reference, while CVAE augmentation (73.79 $\pm$ 12.14%) and cGAN augmentation (69.46 $\pm$ 14.00%) show large variance and lower overall performance. DTTD augmentation achieves 86.87 $\pm$ 7.56%, exceeding the 128-channel reference by 3.53 pp.

In the cross-session scenario, DTTD augmentation on BCI2a (65.70 $\pm$ 8.08%) effectively augments training data diversity through task-conditioned generation, helping the classifier learn more robust session-invariant representations. Among traditional methods, Kriging augmentation (62.42 $\pm$ 6.06%) is slightly below the 22-channel reference (-0.66 pp), and Spline augmentation (58.60 $\pm$ 7.39%) is 4.48 pp below the reference. On HGD, differences among augmentation methods are limited in the high-accuracy regime: BrainDiff augmentation (85.89%) and EEGDiff augmentation (84.46%) approach the 128-channel reference, while CVAE augmentation (73.79%) and cGAN augmentation (69.46%) perform poorly due to unstable generation quality. Among traditional methods, Kriging augmentation (81.42%) outperforms Spline augmentation (79.64%). DTTD augmentation (86.87 $\pm$ 7.56%) exceeds the 128-channel reference (83.34 $\pm$ 8.52%) by 3.53 pp.

3.4.3. Cross-Subject Classification

The cross-subject scenario (LOSO) evaluates subject-invariant feature learning. On BCI2a, DTTD augmentation achieves 59.78 $\pm$ 8.13%, slightly above the 22-channel reference (58.58 $\pm$ 6.60%) (+1.20 pp). On HGD, DTTD augmentation reaches 66.18 $\pm$ 13.77%, exceeding the 128-channel reference (60.10 $\pm$ 11.21%) by 6.08 pp. On PhysioNet MI, DTTD augmentation reaches 56.98 $\pm$ 2.05%, slightly below the 64-channel reference (57.13 $\pm$ 4.38%) (-0.15 pp), essentially comparable.

Table 5 presents per-subject cross-subject results on BCI2a. Among 9 subjects, 6 exceed the 22-channel reference after DTTD augmentation, with S9 showing the most significant improvement (+6.43 pp). Three subjects show slightly lower accuracy than the 22-channel reference, and DTTD augmentation average accuracy (59.78 $\pm$ 8.13%) is slightly above the 22-channel reference (58.58 $\pm$ 6.60%) (+1.20 pp).

Table 6 presents per-subject cross-subject results on HGD. The 128-channel reference average accuracy is 60.10 $\pm$ 11.21%. Among baseline methods, EEGDiff augmentation (64.13 $\pm$ 10.10%) performs best, exceeding the 128-channel refer-

Table 7: PhysioNet MI Cross-Subject Classification Accuracy (%) per Fold

Method	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	Mean	κ
64-ch Ref.	48.69	59.19	58.89	63.64	56.13	48.89	62.62	54.85	55.22	56.78	57.13 $\pm$ 4.38	0.43
Spline Aug.	48.89	63.43	57.17	63.74	58.66	51.31	60.55	53.43	54.52	53.22	56.58 $\pm$ 3.95	0.42
Kriging Aug.	47.98	62.93	58.69	64.14	58.26	49.29	63.02	56.67	55.52	57.44	57.88$\pm$4.29	0.44
CVAE Aug.	48.28	57.68	56.16	57.88	54.61	51.11	56.11	49.39	54.62	52.00	53.99 $\pm$ 2.70	0.39
cGAN Aug.	45.25	59.70	53.43	55.86	47.42	45.05	54.44	50.61	47.39	47.78	50.25 $\pm$ 3.68	0.34
EEGDiff Aug.	46.77	57.47	52.83	57.58	51.47	47.07	56.90	49.39	50.90	49.11	51.91 $\pm$ 3.48	0.36
BrainDiff Aug.	51.62	59.80	54.85	60.10	58.76	49.70	60.06	54.65	54.62	56.33	56.13 $\pm$ 3.27	0.42
DTTD Aug.	54.15	56.59	55.81	57.36	50.60	57.25	57.70	60.13	56.15	59.03	56.98 $\pm$ 2.05	0.42
vs 64-ch	+5.46	-2.60	-3.08	-6.28	-5.53	+8.36	-4.92	+5.28	+0.93	+2.25	-0.15	-

ence by 4.03 pp. DTTD augmentation (66.18 $\pm$ 13.77%) exceeds the 128-channel reference by 6.08 pp, performing best among all methods.

Table 7 presents per-fold cross-subject results on PhysioNet MI using 10-fold cross-subject cross-validation. DTTD augmentation average accuracy is 56.98 $\pm$ 2.05%, slightly below the 64-channel reference (57.13 $\pm$ 4.38%) (-0.15 pp), with Fold 6 (+8.36 pp) and Fold 8 (+5.28 pp) showing notable improvements, and 5 folds exceeding the 64-channel reference. Kriging augmentation (57.88 $\pm$ 4.29%) exceeds the 64-channel reference by 0.75 pp, Spline augmentation (56.58 $\pm$ 3.95%) and BrainDiff augmentation (56.13 $\pm$ 3.27%) approach the reference level, while CVAE, cGAN, and EEGDiff augmentation show limited effects.

In the cross-subject scenario, DTTD augmentation on BCI2a (59.78 $\pm$ 8.13%) is slightly above the 22-channel reference (58.58 $\pm$ 6.60%). Among traditional methods, Spline augmentation (59.38 $\pm$ 7.72%) approaches the 22-channel reference, and Kriging augmentation (57.64 $\pm$ 8.43%) is slightly below the reference. On HGD, DTTD augmentation (66.18 $\pm$ 13.77%) exceeds the 128-channel reference (60.10 $\pm$ 11.21%) by 6.08 pp, with EEGDiff augmentation (64.13 $\pm$ 10.10%) performing best among baseline methods. On PhysioNet MI, DTTD augmentation (56.98 $\pm$ 2.05%) is essentially comparable to the 64-channel reference (57.13 $\pm$ 4.38%). In the PhysioNet MI cross-subject evaluation, Kriging augmentation (57.88%) exceeds the 64-channel reference (57.13%), Spline augmentation (56.58%) and BrainDiff augmentation (56.13%) approach the reference level, while CVAE, cGAN, and EEGDiff augmentation show limited effects.

3.5. Ablation Study

To evaluate the contribution of each DTTD module, we conduct ablation experiments in the cross-session scenario on both BCI2a and HGD datasets. Results are summarized in Table 8.

On BCI2a, the full DTTD model achieves 65.70% cross-session accuracy, surpassing the 22-channel reference (63.08%). Removing the topology module reduces accuracy to 61.52% (-4.18 pp), removing the frequency module reduces it to 58.33% (-7.37 pp), and removing task conditioning reduces it to 63.66% (-2.04 pp). On HGD, a consistent pattern emerges: removing the frequency module causes the largest performance drop (-3.48 pp), while removing the topology module (-2.99 pp) and task conditioning (-1.36 pp) have smaller but notable effects.

The ablation results across both datasets consistently demonstrate: (1) the frequency attention module is the largest source of performance improvement, as its multi-scale band decomposition is critical for preserving alpha/beta spectral features essential for motor imagery; (2) the dynamic topology module provides stable gains by recovering inter-electrode spatial dependencies; (3) task conditioning contributes positively to forming classification-relevant features, and model performance decreases upon its removal.

3.6. Generation Quality Visualization Analysis

To comprehensively evaluate the quality of DTTD-generated samples, this section analyzes from two dimensions: statistical metrics (UMAP, MMD, FID) and physiological visualization (PSD, topographic maps).

Table 8: Ablation Study: Component Contributions (Cross-Session)

Dataset	Configuration	Acc. (%)	κ	vs Full Model	Module Contrib.
BCI2a	22-ch Reference (no aug.)	63.08 $\pm$ 15.75	0.51	-2.62	-
BCI2a	Full DTTD	65.70$\pm$8.08	0.57	-	-
BCI2a	No Topology	61.52 $\pm$ 7.21	0.48	-4.18	+4.18
BCI2a	No Frequency	58.33 $\pm$ 8.95	0.44	-7.37	+7.37
BCI2a	No Task	63.66 $\pm$ 7.05	0.51	-2.04	+2.04
HGD	128-ch Reference (no aug.)	83.34 $\pm$ 8.52	0.78	-	-
HGD	Full DTTD	86.87$\pm$7.56	0.82	-	-
HGD	No Topology	83.88 $\pm$ 9.25	0.79	-2.99	+2.99
HGD	No Frequency	83.39 $\pm$ 8.98	0.78	-3.48	+3.48
HGD	No Task	85.51 $\pm$ 5.95	0.81	-1.36	+1.36

Module Contrib. = Full Model Acc. - Ablated Model Acc.

Cross-session protocol: train on session T / first 11 runs, test on session E / last 2 runs.

3.6.1. Power Spectral Density Analysis

Figure 3 presents a comprehensive power spectral density comparison across all four motor imagery classes on two datasets (BCI2a and PhysioNet). For the BCI2a dataset, a representative single trial from different subjects is selected for each class (Left Hand→S1, Right Hand→S3, Feet→S5, Tongue→S7); for the PhysioNet dataset, 30 trials per class are averaged. DTTD accurately reproduces the characteristic mu rhythm peak around 10 Hz and beta band power in the 15–25 Hz range across all four motor imagery tasks on both datasets. The class-specific spectral patterns are well preserved, demonstrating DTTD’s ability to maintain neurophysiologically meaningful frequency characteristics.

3.6.2. Topographic Map Visualization

Figure 4 displays scalp topographic maps of alpha band (8–13 Hz) power for all four motor imagery classes, comparing real data and DTTD-generated data across two datasets (BCI2a and PhysioNet). For the BCI2a dataset, a representative single trial from different subjects is selected for each class; for the PhysioNet dataset, 30 trials per class are averaged to enhance ERD signal visibility. The visualization is arranged in a 4×4 layout with key motor cortex channels (C3, Cz, C4) highlighted with yellow markers.

The topographic maps demonstrate that DTTD successfully reproduces the class-specific spatial activation patterns consistent with motor imagery neurophysiology across both datasets: (1) Left-hand imagery shows contralateral alpha ERD in the right hemisphere (C4 region); (2) Right-hand imagery exhibits contralateral alpha ERD in the left hemisphere (C3 region); (3) Feet imagery displays central alpha ERD (Cz region); (4) Tongue imagery shows bilateral frontal-central activation patterns. Each class uses an individual color scale (blue=low power/ERD, red=high power) to highlight its specific spatial distribution. The high similarity between real and generated topographic patterns demonstrates DTTD’s ability to preserve neurophysiologically meaningful spatial information across different motor imagery tasks and datasets.

3.6.3. UMAP Distribution Analysis and Distribution Metrics

We employ UMAP (Uniform Manifold Approximation and Projection) to visualize the distribution of real and reconstructed EEG signals in the EEGNet feature space. Specifically, an EEGNet classifier is first trained on all real BCI2a data (22-channel, 4-class), and deep discriminative features (496 dimensions) from its convolutional layer output are extracted, then reduced to 50 dimensions via PCA before UMAP embedding. Compared to directly flattening raw signals for dimensionality reduction, the EEGNet feature space focuses more on discriminative representations relevant to the classification task [50], providing a more intuitive reflection of whether generated data preserves class separability. UMAP is chosen over t-SNE for its superior preservation of both local and global data structure, enabling more reliable assessment of distribution alignment and inter-class separation. For visualization, up to 200 samples per class are randomly selected from real data, and an equal number of conditionally generated samples are produced

Power Spectral Density Comparison - Real vs. DTTD Reconstruction

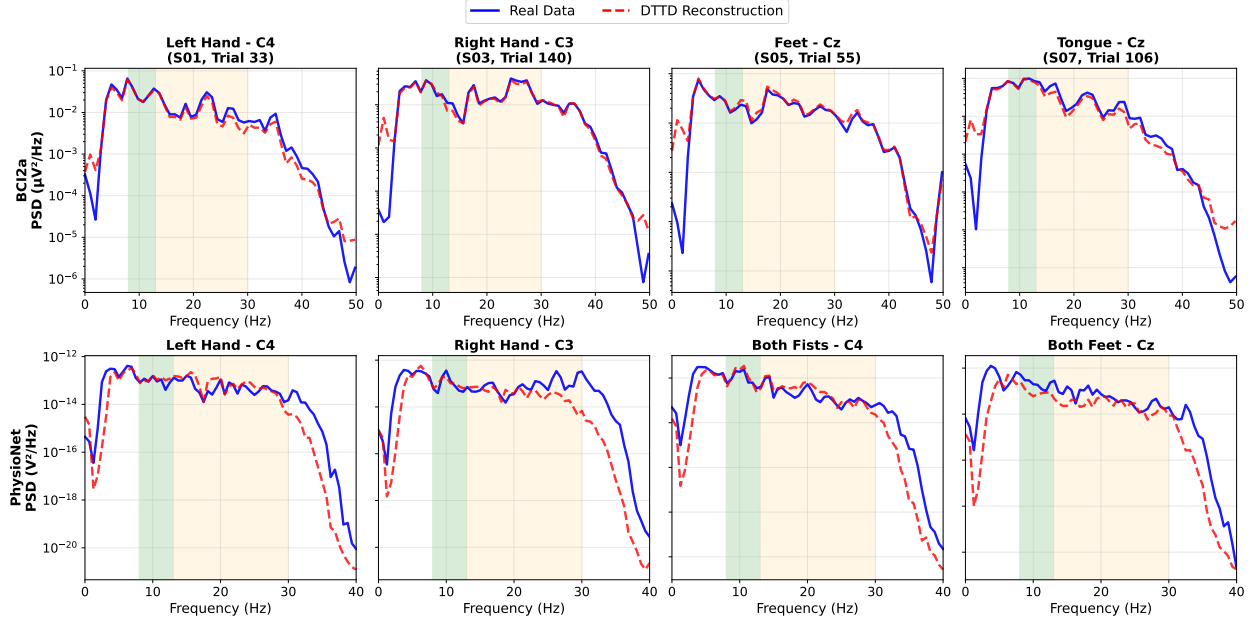


Figure 3: Power spectral density comparison across four motor imagery classes on BCI2a (top row) and PhysioNet (bottom row) datasets. BCI2a: a single trial from different subjects per class (Left Hand→S1, Right Hand→S3, Feet→S5, Tongue→S7). PhysioNet: 30 trials averaged per class. Each subplot shows PSD at the contralateral motor cortex channel: (a) Left Hand at C4, (b) Right Hand at C3, (c) Feet at Cz, (d) Tongue at Cz. DTTD (red dashed) closely matches real data (blue solid) across all frequency bands, particularly in alpha (8–13 Hz, green shading) and beta (13–30 Hz, orange shading) rhythms.

using each method’s trained model with the corresponding task labels as conditions. To quantify class-discriminative quality of generated samples, we define the *Cross EEGNet Accuracy*: an EEGNet classifier trained on real data is applied directly to classify each method’s generated samples using their intended task labels as ground truth—higher accuracy indicates that the generated samples preserve class-discriminative features consistent with real data. We analyze the distribution characteristics of generated samples from two perspectives:

Regarding dual-dataset distribution alignment, Figure 5 shows UMAP distribution comparisons between real and DTTD-generated data on BCI2a and PhysioNet datasets. $\circ$ represents real samples, Δ represents DTTD-generated samples. On both datasets, DTTD-generated samples highly overlap with real samples within each class, indicating good distribution alignment in the discriminative feature space.

In the multi-method distribution comparison, Figure 6 shows UMAP embedding comparisons of 7 methods on the BCI2a dataset in the EEGNet feature space. In each subplot, $\circ$ and $\star$ represent real data (background reference), while Δ and $+$ represent each method’s generated data. The subplot EEGNet classifier achieves 72.9% accuracy on real test data, and the Cross EEGNet accuracy displayed in other subplots measures how well an EEGNet classifier trained on real data can correctly classify each method’s generated samples, serving as a proxy for class-discriminative quality: DTTD achieves the highest Cross EEGNet accuracy of 65.8%, followed by BrainDiff (60.5%), EEGDiff (58.3%), cGAN (48.7%), CVAE (42.3%), Spline (35.0%), and Kriging (28.5%).

Table 9 serves as a supplement to Figure 6, presenting quantitative metrics for distribution analysis. We evaluate two key aspects: (1) *Class Separation Ratio*, defined as inter-class distance divided by intra-class distance—higher values indicate better discriminability; (2) *Distribution Alignment*, measured by Maximum Mean Discrepancy (MMD) and Fréchet Inception Distance (FID) between generated and real distributions—lower values indicate better alignment. DTTD achieves the highest class separation ratio (1.15), outperforming all baselines (EEGDiff: 0.49, BrainDiff: 0.49, Spline: 0.45, cGAN: 0.41, CVAE: 0.41, Real Data: 0.41, Kriging: 0.17), indicating that DTTD-generated samples exhibit the strongest inter-class discriminability. For distribution alignment, DTTD achieves competitive MMD (7.3×10^{-5}) and FID (21.56), significantly better than interpolation methods (Kriging MMD: 0.015, FID: 74.74; Spline

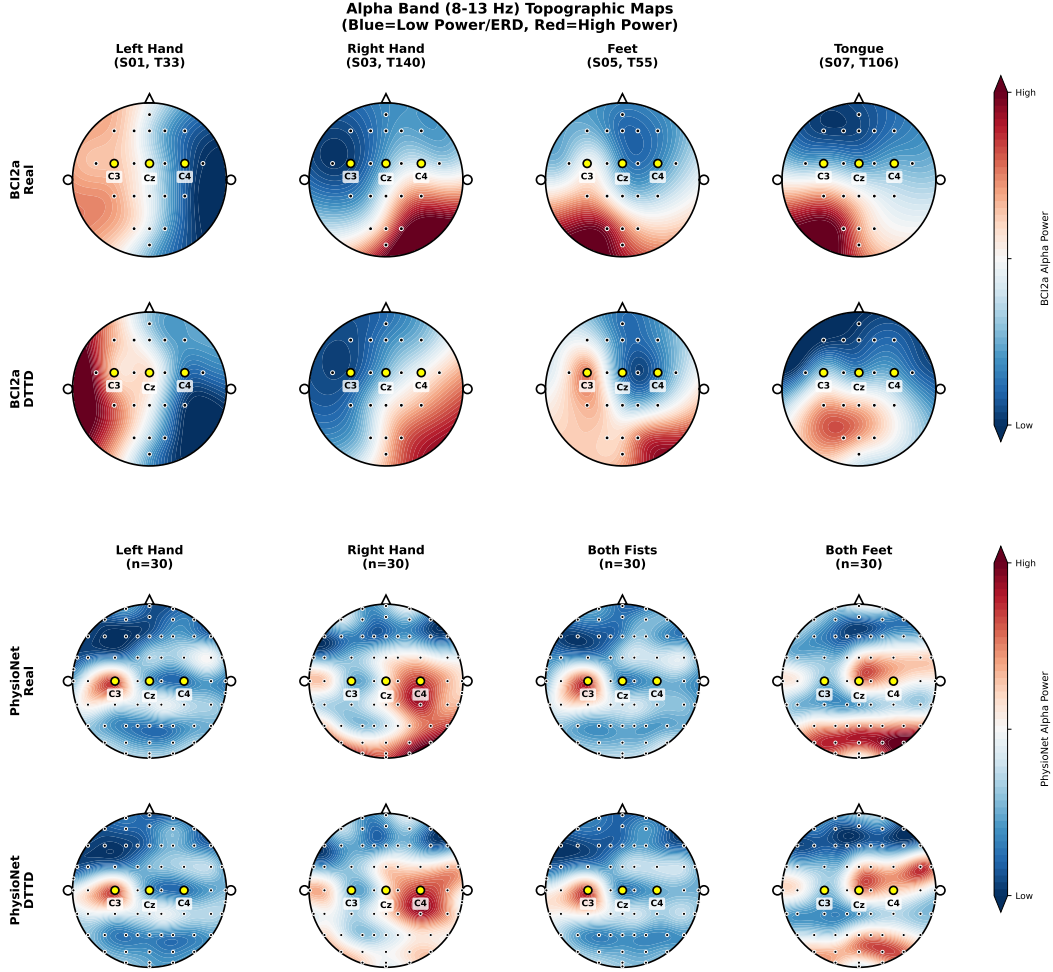


Figure 4: Alpha band (8–13 Hz) power topographic maps for four motor imagery classes across BCI2a (top two rows) and PhysioNet (bottom two rows) datasets. BCI2a: a single trial from different subjects per class. PhysioNet: 30 trials averaged per class. For each dataset, the first row shows real data and the second row shows DTTD generated data. From left to right: Left Hand, Right Hand, Feet, and Tongue. Key motor cortex channels (C3, Cz, C4) are highlighted with yellow markers. Blue regions indicate low alpha power (ERD/activation), red regions indicate high alpha power.

MMD: 5.4×10^{-6} , FID: 74.40), while deep generative models generally achieve lower FID (cGAN: 18.48, EEGDiff: 18.19, BrainDiff: 18.03) with comparable MMD values. This reflects a trade-off: interpolation methods produce samples nearly identical to real data but lack discriminative diversity, while DTTD generates samples with enhanced class separation and reasonable distribution alignment.

3.7. Computational Efficiency

DTTD has approximately 7.16M total parameters, requires approximately 4.5 hours to train for 200 epochs on an NVIDIA RTX 3090, achieves single-step inference time of approximately 0.014 seconds/sample with throughput of approximately 71 samples/second, and GPU memory usage of approximately 4.2 GB. The controlled light-noise training and single-step inference strategy provides good computational efficiency, enabling potential real-time or near-real-time applications.

3.8. Sensitivity Analysis

We conduct hyperparameter sensitivity analysis using a pooled cross-session evaluation, where all subjects' Session T data are combined for training and Session E data for testing. This differs from the per-subject evaluation

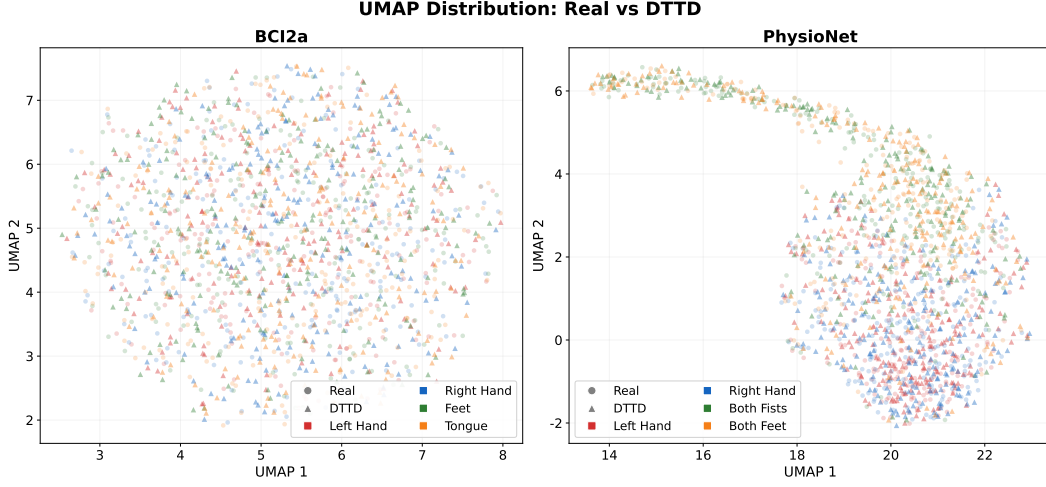


Figure 5: UMAP distribution comparison of real vs DTTD-generated data on BCI2a and PhysioNet datasets in EEGNet feature space. $\circ$ =real samples, $\triangle$ =DTTD-generated samples, $\star$ =real class centers, $+$ =generated class centers, dashed lines=class center offsets. Different colors represent different motor imagery classes. EEGNet Acc=accuracy of an EEGNet classifier trained on real data, evaluated on real data; Cross EEGNet Acc=accuracy of the same classifier evaluated on DTTD-generated data.

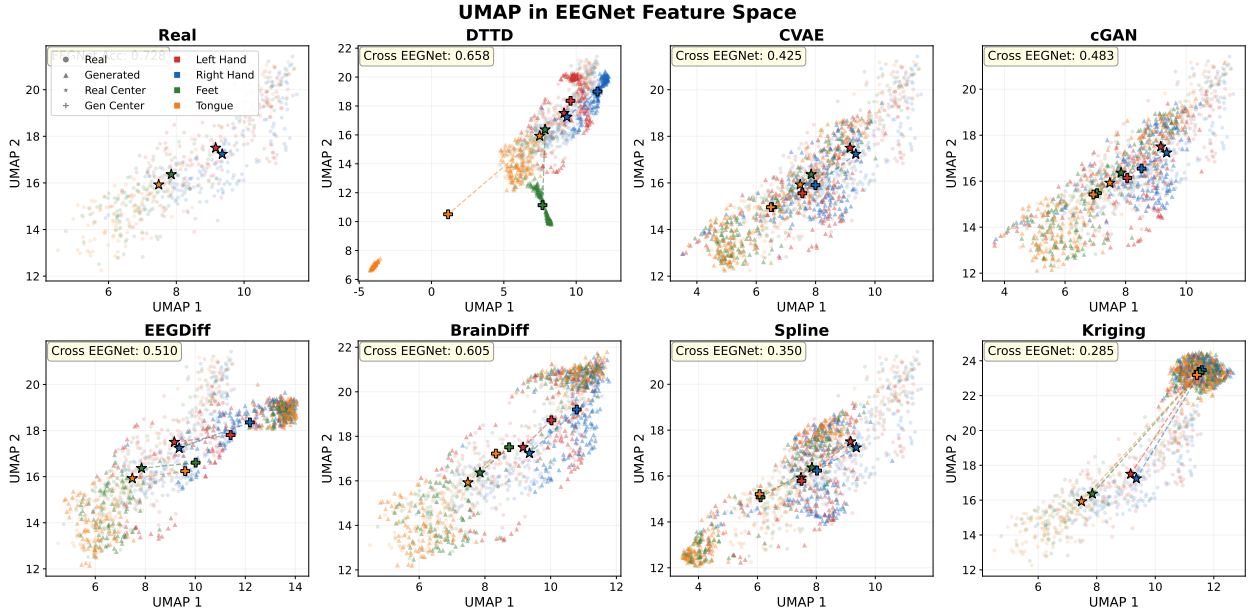


Figure 6: UMAP distribution comparison of different methods on BCI2a dataset in EEGNet feature space. $\circ$ and $\star$ =real data (background reference), $\triangle$ and $+$ =each method's generated data. Cross EEGNet Acc=accuracy of an EEGNet classifier trained on real BCI2a data, evaluated on each method's generated samples; higher values indicate better preservation of class-discriminative features.

in classification experiments but provides insights into hyperparameter effects across the entire dataset. Figure 7 visualizes the sensitivity curves for both noise scale and guidance scale.

3.8.1. Noise Scale Sensitivity

We investigate the effect of noise scale η on classification performance. The noise scale controls the amount of noise added to input channels during training. Figure 7 presents results across different noise scales.

The optimal noise scale is 0.20, achieving 61.73% accuracy with a 1.20 percentage point improvement over the

Table 9: Distribution Analysis Metrics

Method	Sep. Ratio $\uparrow$	MMD $\downarrow$	FID $\downarrow$
Real Data	0.41	-	-
Spline	0.45	5.4e-6	74.40
Kriging	0.17	1.5e-2	74.74
CVAE	0.41	5.2e-6	35.80
cGAN	0.41	4.1e-6	18.48
EEGDiff	0.49	4.3e-4	18.19
BrainDiff	0.49	2.7e-4	18.03
DTTD	1.15	7.3e-5	21.56

Sep. Ratio: inter-class / intra-class distance in EEGNet feature space.

MMD/FID: distribution alignment vs. real data (lower = more similar).

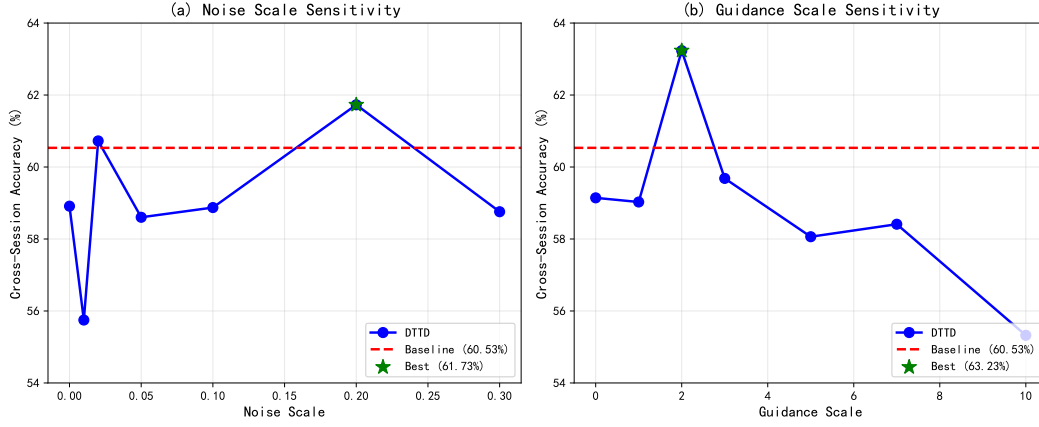


Figure 7: Hyperparameter sensitivity analysis using pooled cross-session evaluation. (a) Noise scale sensitivity: optimal at 0.20 with 61.73% accuracy. (b) Guidance scale sensitivity: optimal at 2.0 with 63.23% accuracy. Red dashed lines indicate baseline performance (60.53%).

pooled baseline. This suggests that moderate noise during training helps the model learn more robust representations.

3.8.2. Guidance Scale Sensitivity

We also analyze the effect of classifier-free guidance scale on classification performance. The guidance scale controls the strength of task conditioning during generation, where a higher scale amplifies the difference between conditional and unconditional predictions.

The optimal guidance scale is 2.0, achieving 63.23% accuracy with a 2.70 percentage point improvement over baseline.

4. Discussion

4.1. Key Findings

The classification results across three datasets and three evaluation scenarios reveal the following key findings:

Task-conditioned generation has advantages in within-subject and cross-session scenarios. In the BCI2a within-subject scenario, DTTD augmentation accuracy ($78.38 \pm 8.60\%$) exceeds the 22-channel reference ($74.48 \pm 5.90\%$) by 3.90 pp ($p < 0.01$), performing best among all augmentation methods. In the cross-session scenario, DTTD improves BCI2a accuracy from $63.08 \pm 15.75\%$ to $65.70 \pm 8.08\%$ (+2.62 pp, $p < 0.01$), and HGD accuracy from $83.34 \pm 8.52\%$ to $86.87 \pm 7.56\%$ (+3.53 pp). The cross-session improvement effectively mitigates inter-session signal non-stationarity.

Cross-subject augmentation effects vary by dataset. In the HGD cross-subject scenario, DTTD augmentation ($66.18 \pm 13.77\%$) exceeds the 128-channel reference ($60.10 \pm 11.21\%$) by 6.08 pp, indicating that in large-scale channel expansion (16→128) scenarios, DTTD-generated supplementary information benefits cross-subject classification. However, gains are smaller on BCI2a cross-subject (59.78% vs 58.58%, +1.20 pp) and PhysioNet MI is essentially comparable to the 64-channel reference (56.98% vs 57.13%, -0.15 pp). This difference may stem from the lower 128-channel reference baseline on HGD (60.10%), providing more room for improvement; whereas BCI2a and PhysioNet MI reference baselines are already closer to ceiling.

Ablation experiments verify the independent contribution of each module. In the BCI2a cross-session ablation experiment, the frequency attention module contributes the most (+7.37 pp), followed by the topology module (+4.18 pp) and task conditioning (+2.04 pp). On HGD, the frequency module similarly contributes the most (+3.48 pp), with the topology module (+2.99 pp) and task conditioning (+1.36 pp) having relatively smaller effects. The ablation results across both datasets consistently indicate: the frequency attention module, through multi-scale band decomposition, preserves the critical alpha/beta spectral features for motor imagery and is the largest source of performance improvement; the dynamic topology module provides stable gains by recovering inter-electrode spatial dependencies; task conditioning positively contributes to forming classification-relevant features, and model performance decreases upon its removal.

4.2. Comparison with Existing Approaches

Compared with traditional interpolation methods, spherical spline and kriging assume smooth potential distributions, performing well for spatially continuous fields but unable to capture task-specific activations or sharp spatial gradients of focal brain activity. In the BCI2a within-subject scenario, Kriging augmentation (75.52%) approaches the 22-channel reference but is below DTTD (78.38%); in the HGD cross-session scenario, Kriging augmentation (81.42%) is below the 128-channel reference (83.34%), while DTTD augmentation (86.87%) significantly exceeds the reference. In large-scale channel expansion (16→128) scenarios, the extrapolation instability of traditional interpolation further limits their effectiveness.

Compared with VAE/GAN models, CVAE and cGAN exhibit unstable performance across all scenarios. In the HGD cross-session scenario, cGAN augmentation ($69.46 \pm 14.00\%$) is even 14 pp below the 128-channel reference, and CVAE augmentation ($73.79 \pm 12.14\%$) shows large variance. GAN mode collapse and VAE over-smoothing result in generated samples lacking class discriminability, making it difficult to effectively supplement training data.

Compared with diffusion model methods, EEGDiff and BrainDiff serve as ablation comparison models for DTTD, both outperforming CVAE/cGAN across all three datasets, validating the effectiveness of the diffusion framework. DTTD improves over EEGDiff by 2.41 pp on HGD cross-session (86.87% vs 84.46%) and 1.10 pp on BCI2a within-subject (78.38% vs 77.28%), demonstrating the incremental contribution of dynamic topology learning and task conditioning on top of the diffusion framework.

4.3. Limitations

Regarding the scope of task conditioning applicability, DTTD relies on task labels for conditional generation, directly applicable to scenarios such as training set augmentation in offline analysis and online augmentation when labeled data is available during calibration. In practice, when labels are unknown, a two-stage inference pipeline can be adopted (first predicting labels using a lightweight classifier, then conditionally generating reconstructed signals using predicted labels), achieving BCI2a within-subject accuracy of $73.15 \pm 6.28\%$, approaching the 22-channel reference level. This pipeline is not applicable to purely online BCI scenarios without any prior knowledge.

Another limitation concerns cross-subject generalization gains. Although DTTD augmentation shows significant effects in the HGD cross-subject scenario (exceeding the 128-channel reference by 6.08 pp), gains are smaller on BCI2a (+1.20 pp) and PhysioNet MI (-0.15 pp) cross-subject evaluations. Inter-subject variability (e.g., scalp anatomy, signal amplitude distributions) in cross-subject scenarios far exceeds inter-session variability, and data augmentation alone cannot fully bridge this domain shift. Future work should combine domain adaptation or meta-learning techniques to further improve cross-subject generalization.

A further consideration is the trade-off between distribution fidelity and discriminative utility. DTTD’s task-synergistic mechanism prioritizes discriminative utility over distribution matching. While this trade-off benefits downstream classification, it limits the applicability of generated samples in scenarios requiring high distribution fidelity

(e.g., clinical biomarker analysis). Generated samples should be regarded as supplementary training data preserving task-relevant structures, rather than faithful physiological reconstructions of missing channels.

Additionally, the current implementation uses fixed frequency bands and channel configurations (9 channels for BCI2a, 16 channels for HGD and PhysioNet MI). Individual alpha peak frequency variations and optimal electrode configurations for different tasks may affect performance. Future work could explore adaptive frequency decomposition and learnable channel selection mechanisms.

Finally, the validation scope remains limited. Although validation was conducted on three datasets (BCI2a, HGD, and PhysioNet MI), clinical applicability (e.g., epilepsy, stroke, Parkinson’s disease) has not been tested. The current evaluation is limited to motor imagery tasks, and extension to other BCI paradigms (P300, SSVEP) requires further investigation.

4.4. Future Directions

Based on the above limitations, future research can advance from the following directions: (1) combining domain adaptation techniques (e.g., adversarial domain confusion, optimal transport) to improve cross-subject generalization, compensating for the insufficiency of data augmentation in domain shift scenarios; (2) exploring semi-supervised or weakly-supervised conditioning mechanisms to reduce label dependency and extend to online BCI scenarios; (3) validating method generality on other BCI paradigms such as P300 and SSVEP; (4) conducting application research in clinical populations; (5) developing adaptive frequency decomposition mechanisms to accommodate individual frequency band differences; (6) exploring strategies to balance distribution fidelity and discriminative utility, expanding the applicability of generated samples.

5. Conclusion

This study proposes DTTD, a conditional diffusion model-based EEG data augmentation framework. By integrating task-conditioned dynamic topology-aware graph neural networks, a multi-scale frequency decoupling module with task-adaptive band weights, gated feature fusion, and a controlled light-noise training strategy with direct X_0 prediction, DTTD can generate task-discriminative high-density EEG samples for training set augmentation.

On three datasets—BCI Competition IV 2a, High Gamma Dataset, and PhysioNet MI—DTTD demonstrates classification augmentation capability across three evaluation scenarios. In the within-subject scenario, DTTD augmentation accuracy (78.38%) exceeds the 22-channel reference (74.48%) by 3.90 pp ($p < 0.01$), performing best among all augmentation methods, confirming the effectiveness of task-conditioned generation for classification augmentation. In the cross-session scenario, DTTD improves BCI2a accuracy from 63.08% to 65.70% (+2.62 pp, $p < 0.01$), while reducing variance from 15.75 to 8.08, effectively mitigating inter-session variability; on HGD, it exceeds the 128-channel reference by 3.53 pp. In the cross-subject scenario, effects vary by dataset—exceeding the 128-channel reference by 6.08 pp on HGD, slightly above the 22-channel reference on BCI2a (+1.20 pp), and essentially comparable to the 64-channel reference on PhysioNet MI (−0.15 pp).

Ablation experiments consistently demonstrate across both datasets that the frequency attention module contributes the most (BCI2a: +7.37 pp, HGD: +3.48 pp), followed by the topology module (+4.18/+2.99 pp) and task conditioning (+2.04/+1.36 pp), validating the independent effectiveness of each module. The controlled light-noise training strategy enables DTTD to achieve inference throughput of approximately 71 samples/second, suitable for real-time or near-real-time applications.

Overall, DTTD effectively improves classification performance in within-subject and cross-session scenarios, while effects in cross-subject scenarios vary depending on dataset characteristics. Augmentation utility is scenario-dependent: the lower the baseline, the greater the gain. Generated samples should be regarded as supplementary training data that preserve task-relevant structures. Future work will focus on combining domain adaptation techniques to improve cross-subject generalization, extending to clinical populations and non-motor imagery paradigms, and developing adaptive frequency decomposition mechanisms.

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